Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	I Love You 3000
Project / Solution Name	Domain 1 - The Aiversecity Future-ReadyWorkforce Domain
Name (Leader)	GUI GUANYU
Email & Phone (Leader)	noteiinoteii2004@gmail.com & 0129307921
University Affiliation(s)	UNIVERSITI TEKNOLOGI MALAYSIA (UTM)
Live Prototype URL	Custom Domain: https://www.skillverse-ai.site/ Vercel Mirror: https://skill-verse-ai-gray.vercel.app/ Replit Mirror: https://skill-verse-ai.replit.app (You may use any link listed at here.)
Code Repository URL (public / access granted to judges)	https://github.com/njwongzhe/SkillVerse-AI

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

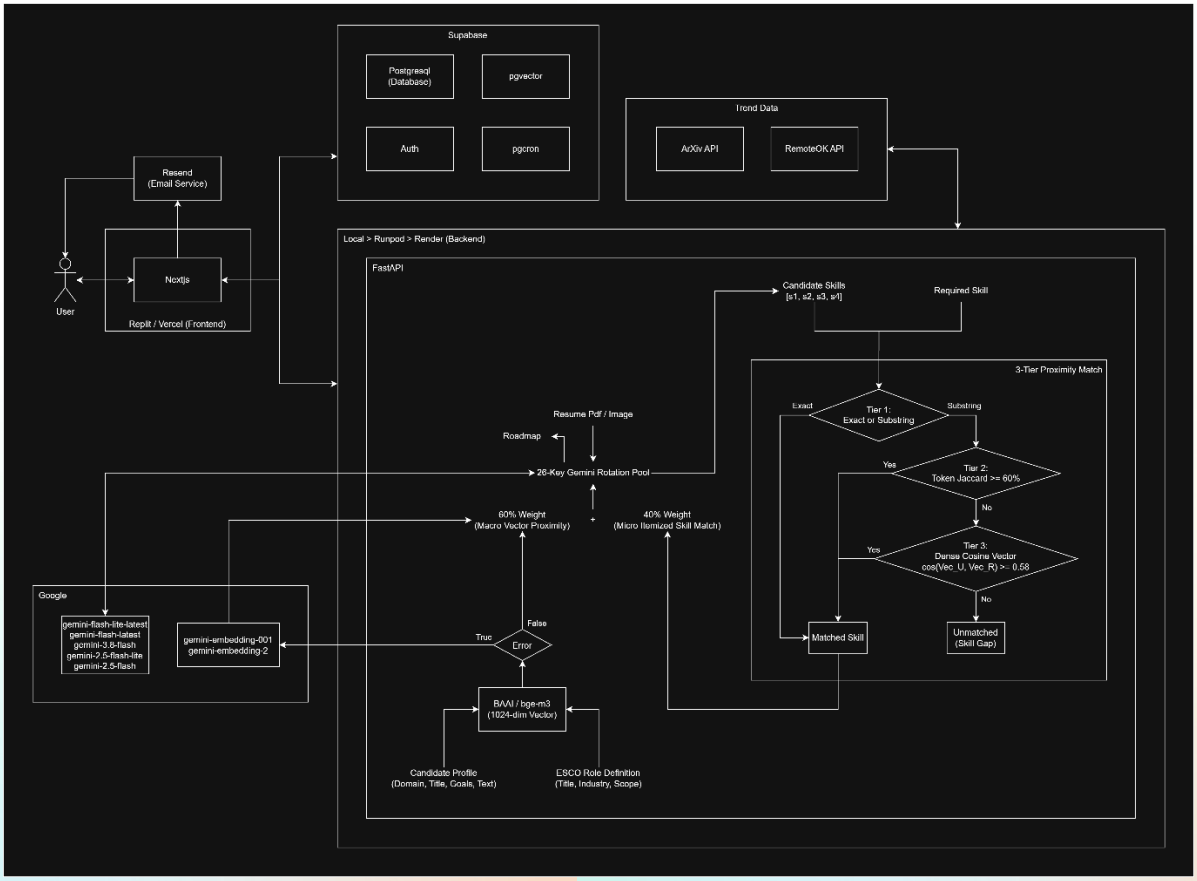
SkillVerse AI connects a decoupled web client, resilient backend gateways, GPU-accelerated vector inference and real-time labor data pipelines.

The user interface, developed in Next.js 16 with React Flow, is dual hosted on Replit for collaborative prototyping and Vercel for Edge failover. Client interactions route through an intelligent Next.js gateway that manages a three-tier failover mechanism across local instances, RunPod GPU proxies and Render hosted web services.

The backend computation engine (FastAPI, Python 3.11) processes multimodal profile inputs via a circular 26-key Gemini rotation pool (5-Model Waterfall) that strips PII for demographic neutrality. Roles and skills are normalized against European ESCO taxonomies persisted in Supabase PostgreSQL 15, accelerated by pgvector HNSW indexing.

Matching executes via a deterministic dual-track formula which is 60% macro semantic similarity computed with BAAI/bge-m3 dense embeddings hosted on RunPod Cloud GPUs, with an identical fallback mirror on Render plus 40% micro itemized overlap via a three-tier proximity algorithm.

Autonomous background workers continuously ingest ArXiv cs.AI and RemoteOK feeds, calculating live skill velocities (V_s) and dynamically expanding unindexed competencies directly into the database.



Link: https://github.com/njwongzhe/SkillVerse-AI/blob/main/docs/Simple%20Project%20Box%20Diagram.png

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.3	Static typing and interface contracts across the client web application.	Apache-2.0
Python 3.11	Asynchronous computational backend programming runtime and service environment.	Python Software Foundation (PSF) License
SQL (PostgreSQL 15)	Relational database schema definition, migrations, and vector queries.	PostgreSQL Open-Source License

Name & Version	Purpose in Project	License
Next.js 16.4.2	Full-stack web client framework, SSR, and gateway failover client.	MIT License
React 19.3.0	Component tree rendering and reactive client state management.	MIT License
React DOM 19.3.0	Document Object Model rendering engine for React web interfaces.	MIT License
FastAPI >=0.115.0	Asynchronous REST API framework, microservice orchestration, and SSE streaming.	MIT License
Uvicorn >=0.30.0	High-performance ASGI web server runtime for FastAPI services.	BSD-3-Clause
Supabase pgvector	Database extension for high-dimensional vector storage and HNSW indexing.	PostgreSQL Open-Source License
Prisma ^7.10.1	Object-Relational Mapping (ORM) and type-safe database queries.	Apache-2.0
@prisma/adapter-pg ^7.10.1	Direct PostgreSQL connection adapter for Prisma client execution.	Apache-2.0
@supabase/supabase-js ^2.118.0	Client SDK for Supabase authentication, database queries, and storage.	MIT License
pg ^8.23.1	Native PostgreSQL driver and connection pooling for Node.js runtime.	MIT License
Tailwind CSS 4.3.3	Utility-first CSS styling framework for responsive web user interfaces.	MIT License
next-intl ^4.14.9	Trilingual internationalization routing and dictionary localization (en, ms, zh).	MIT License
React Flow 11+	Interactive node-based graph rendering for personalized reskilling roadmaps.	MIT License
Lucide React	Visual icon library for dashboard telemetry and UI navigation elements.	MIT License
html2canvas ^1.4.1	Client-side screenshot capture of HTML DOM elements for report exporting.	MIT License

Name & Version	Purpose in Project	License
jsPDF ^4.2.1	Client-side generation and export of cryptographic credential PDF files.	MIT License
simpleheat ^0.4.0	Lightweight HTML5 canvas library for rendering enterprise skill deficit heatmaps.	BSD-2-Clause
PyTorch >=2.0.0	GPU/CPU tensor batch operations and vector distance mathematical calculations.	BSD-3-Clause
transformers >=4.40.0	Deep learning model tokenization and transformer architecture pipeline execution.	Apache-2.0
sentence-transformers >=3.0.0	Framework for high-throughput sentence, text, and code dense embeddings.	Apache-2.0
fastembed >=0.3.0	Lightweight, fast embedding inference engine for local vector generation.	Apache-2.0
google-genai >=1.0.0	Official Python client library for accessing Google Gemini GenAI APIs.	Apache-2.0
Pydantic >=2.9.0	Strict data validation, schema enforcement, and payload parsing	MIT License
pydantic-settings >=2.5.0	Environment variable loading and configuration management for FastAPI	MIT License
pypdf >=5.0.0	Native unencrypted PDF text parsing fast-path to bypass vision OCR.	BSD-3-Clause
Pillow >=10.4.0	Image processing library for handling multimodal CV scans and credentials.	MIT-CMU
httpx >=0.27.0	Asynchronous HTTP client for fetching external labor trends and remote APIs.	BSD-3-Clause
python-dotenv >=1.0.1	Reads key-value pairs from .env files into system environment variables.	BSD-3-Clause
python-multipart >=0.0.9	Streaming parser for handling multipart/form-data CV file uploads.	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API (Google)	Multimodal CV extraction, PII scrubbing, contextual AI mentor, and quiz synthesis.	Free Tier
Google Gemini Embeddings API (Google)	Cloud vector fallback cascade tiers for semantic similarity embedding.	Free Tier
Supabase Data API (Supabase)	Relational database operations and pgvector HNSW similarity queries.	Free Tier
Supabase Auth API (Supabase)	User authentication, token issuance, and account session management.	Free Tier
ArXiv API (export.arxiv.org) (Cornell University)	Automated ingestion of live AI research abstracts (cs.AI) for skill velocity (V_s).	Free Tier / Public Open API
RemoteOK API (remoteok.com/api) (RemoteOK)	Live job listing ingestion to compute real-time market demand velocities (V_s).	Free Tier / Public Open API
RunPod Inference API (RunPod)	Remote GPU tensor execution endpoint for BAAI/bge-m3 batch vector calculations	Sponsor-Provided
Render Web Service API (Render)	Mirrored secondary backend web service (Hot standby failover host).	Free Tier
Resend Email API (Resend)	Transactional email delivery for account creation and milestone notification.	Free Tier

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Platform (Replit)	SDC Sponsor platform for collaborative prototyping and live frontend evaluation.	Sponsor-Provided / Global

Service & Provider	Purpose in Project	Plan / Region
Vercel Platform (Vercel)	Standalone Edge production hosting for Next.js frontend (Failover backup node).	Free Tier / Global
RunPod Cloud Platform (RunPod)	Primary backend compute and GPU acceleration for BAAI/bge-m3 tensor vectorization.	Sponsor-Provided / CA-MTL-1
Render Cloud Platform (Render)	Secondary mirrored backend instance (Hot standby failover if RunPod is unreachable).	Free Tier / Singapore
Supabase PostgreSQL (Supabase)	Managed PostgreSQL 15 database hosting pgvector, ESCO roles, and trend signals	Free Tier / ap-southeast-2
Docker Engine & Runtime	Local containerization and standardized production container builds.	Community Edition / Local & Cloud Runtime

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gemini-flash-lite-latest	Google AI Studio	Pre-trained (Primary fast model for PII redaction and CV parsing.)	Google GenAI Terms of Service
gemini-flash-latest	Google AI Studio	Pre-trained (Contextual AI mentor and adaptive pathway synthesis.)	Google GenAI Terms of Service
gemini-3.8-flash	Google AI Studio	Pre-trained (Dynamic novel skill concept synthesis and milestone quizzes.)	Google GenAI Terms of Service
gemini-2.5-flash-lite	Google AI Studio	Pre-trained (Waterfall failover tier 4 to mitigate HTTP 429 rate limits.)	Google GenAI Terms of Service
gemini-2.5-flash	Google AI Studio	Pre-trained (Waterfall failover tier 5 to mitigate HTTP 429/503 errors.)	Google GenAI Terms of Service
gemini-embedding-001	Google AI Studio	Pre-trained (Cloud embedding fallback cascade tier 3.)	Google GenAI Terms of Service
gemini-embedding-2	Google AI Studio	Pre-trained (Cloud embedding fallback cascade tier 4.)	Google GenAI Terms of Service
BAAI/bge-m3 (1024-dim)	Hugging Face / BAAI	Pre-trained (Core dense multilingual vector model for macro role matching.)	MIT License

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
European ESCO Occupational Taxonomy	European Commission Open Data (ec.europa.eu/esco)	Public Open Dataset (104+ curated standard roles & competencies.)	Open Data Commons (ODC-BY 1.0)	N
UN SDG Targets & Indicators Taxonomy	United Nations Department of Economic & Social Affairs	Public Open Dataset (Official definitions for 17 SDG goals & targets.)	Public Domain / CC BY 4.0	N
ArXiv cs.AI Research Feed	export.arxiv.org/api	Public Open Data Stream (Live AI research abstracts for trend velocity.)	CC0 / Public Domain	N
RemoteOK Live Labor Signals	remoteok.com/api	Public Open Data Stream (Live remote job postings for skill demand.)	Creative Commons Attribution	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Antigravity	Primary programming engine for full-stack feature implementation, architectural refactoring and automated test suite creation / execution.	Major
Replit (AI Environment & Agent)	Assisting with cloud workspace configuration, automated frontend deployment scripts, and generating project slide decks.	Moderate
Google Gemini (Web)	Idea brainstorming, technical architecture evaluation, and recommending / explaining mathematical formulas and algorithms.	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

While Antigravity handled core programming and test scaffolding, our engineering team manually reviewed every code commit, verified business logic and validated unit and integration tests. Architectural ideas and algorithm formulas recommended by Gemini were verified against ESCO standards and rigorous mathematical proofs before implementation. Replit deployment routines and presentation slides were audited for operational stability and factual precision. The team maintains complete ownership and technical accountability for all deployed code, system outputs and project materials.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Platform / Cloud Dev Environment & Hosting	Used for collaborative development, configuring deployment routines for live frontend preview and generating presentation slide decks.
RunPod	Cloud GPU Compute Credits / Platform	Hosted high-throughput GPU Pod instances executing BAAI/bge-m3 batch tensor calculations for dense vector matching.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Sponsor resources greatly accelerated prototyping, but our architecture remains resilient and modular. Frontend deployment on Replit is mirrored on an independent Vercel instance to safeguard against credit exhaustion. RunPod GPU compute substantially cuts vector latency. However, our 3-tier gateway failover automatically routes requests to Render cloud compute or local PyTorch inference if RunPod becomes unavailable. Consequently, the prototype operates reliably across both sponsor-backed and standalone production environments.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

SkillVerse AI strictly adheres to zero-retention and data-minimization principles. The platform does not collect, log or persist personally identifiable information (PII).

When a candidate uploads a CV (PDF, image, or text), it passes immediately through our Service 1 PII-Scrubbing engine. Sensitive attributes which including full names, government IDs, contact credentials, physical addresses, age, gender, ethnicity and photos are stripped in volatile memory before reaching vector models or LLM APIs. Only anonymized competency records and normalized domain experience tokens are processed.

Session states and temporary evaluation metrics are stored with ephemeral session IDs in Supabase and can be wiped immediately upon user request. No user resumes or profile records are committed to persistent storage or used for model training, ensuring complete demographic neutrality and data privacy.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Primary technical risks include API rate limits, model hallucinations, external network volatility and third-party platform dependencies.

To sustain production-grade throughput under zero-budget constraints without commercial subscription overhead, we engineered a circular 26-key rotation pool across a 5-model Gemini waterfall, fully resolving free-tier RPM/TPM restrictions (HTTP 429) via automated multi-tier load distribution.

To eliminate career matching hallucinations, match scoring completely bypasses generative models; it is governed by a deterministic mathematical formula $(0.60 \times \text{Macro} + 0.40 \times \text{Micro})$ validated against verified ESCO taxonomies.

To mitigate vendor lock-in, our 4-tier embedding cascade automatically falls back from GPU-hosted BAAI/bge-m3 to local CPU or Gemini embeddings.

Finally, our 3-tier gateway failover seamlessly routes traffic from local nodes through RunPod to Render, and our web client mirrors across Replit and Vercel, ensuring continuous availability even if primary cloud providers experience downtime.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



GUI GUANYU

23/10/2026